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ABSTRACT

Purpose: This study evaluated how extended match time (90+30 min) affected physiological responses and fatigue in male soccer players. **Methods:** Twenty competitive players (age 21±2 [±SD] years; VO_{2max} 59±4 ml/min/kg) completed an experimental match with their activity pattern and HR assessed throughout the game, while countermovement jump (CMJ) performance and repeated sprint ability (RSA) were tested and quadriceps muscle biopsies and venous blood samples taken at baseline and after 90 and 120 min of match-play. **Results:** Less high-intensity running (12%) was performed in extra time in association with fewer intense accelerations and decelerations per minute compared to normal time. Peak sprint speed was 11% lower in extra time compared to normal time, and fatigue also manifested in impaired post-match RSA and CMJ performance (all $p<0.05$). Muscle glycogen declined from 373±59 at baseline to 266±64 $mmol \cdot kg^{-1}$ d.w. after 90 min, with a further decline to 186±56 $mmol \cdot kg^{-1}$ d.w. following extra time ($p<0.05$) and with single fiber analyses revealing depleted or very low glycogen levels in ~75% of both slow and fast twitch fibers. Blood glucose did not change during the first 90-min but declined ($p<0.05$) to 81±8 mg·dL after extra time. Plasma glycerol and ammonia peaked at 236±33 mg·dL and 75±21 μmol·L after the extra period. **Conclusions:** These findings demonstrate exacerbated fatigue following extra time compared with normal time, which appears to be associated with muscle glycogen depletion, reductions in blood glucose levels and hyperammonemia. Together, this points to metabolic disturbances being a major part of the integrated and multifaceted fatigue response during extended soccer match-play.

Key Words: FOOTBALL OVERTIME, MUSCLE GLYCOGEN, FATIGUE, PERFORMANCE, CENTRAL FATIGUE, HYPERAMMONEMIA

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INTRODUCTION

The intensity and derived physiological demands in international competitive male soccer have increased markedly over the last 50 years, and this trend is expected to continue in the upcoming decade (1). In addition to intensified matches, more frequent participation in European club tournaments and national team tournaments is contributing to a further acceleration of the overall physiological demands of modern male soccer, as will the proposed expansion of the number of teams in future World Cups (1). In this context, extended match time is an important but often ignored factor, with very little scientific evaluation of its importance for fatigue development and performance (fatigue herein defined as a disabling symptom in which physical and cognitive function is limited by interactions between performance fatigability and perceived fatigability (2)).

At the men´s UEFA EURO 2020 tournament, more than half of the games in the knock-out stages went to extra time (8 of 15) and more than 85% of the medalists in the men´s EUROs and World Cups since 1992 have participated in games going to extra time. Thus, extended game scenarios frequently occur and are almost mandatory in order to go through to final cup stages in modern tournaments. The establishment of a new European club tournament, i.e. the UEFA Europa Conference League, the addition of the Nations League for national teams and the decision to expand the number of teams in future World Cups are likely to increase the occurrence of 120-min soccer games. Furthermore, the recent rule change about abolishing the advantage of away goals in international and national soccer will potentially contribute to this development and further increase the frequency of games going into extra-time.

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It is well established that fatigue develops during an elite soccer game and peaks in the last 15 min, which compromises high-intensity performance (3). Undeniably, repeated sprint ability (4, 5), muscle strength (6) and muscle function (7) are markedly impaired after a normal game. Muscle physiological mechanisms leading to muscle fatigue towards the end of a game are likely to be multifactorial, but may in part originate from altered muscle metabolism and partially depleted muscle glycogen stores (8). Indeed, it has been reported in male (9) and female (5) players that a substantial fraction of both slow twitch (ST) and fast twitch (FT) fibres are fully or nearly depleted of glycogen after a normal game. Likewise, in both of the aforementioned studies correlations were found between the drop in muscle glycogen and impairment in post-game repeated sprint performance (5, 9). Thus, pronounced muscle glycogen degradation during a soccer game is likely to be associated with end-game fatigue, especially if the depletion of single fibers becomes pronounced and the mixed muscle glycogen content decline to 250 $mmol \cdot kg^{-1}$ d.w. or below as this has been associated with impaired muscle function (10). Specifically, it has been demonstrated that glycogen stored in subcellular pools in direct association with the main steps in the excitation-contraction coupling (e.g. Ca^{2+} handling, Na^{+} and K^{+} regulation, and crossbridge cycling (10) is affected already as consequence of an ordinary 90 min soccer game (11), and this impairment of muscle excitability and contractility could be further aggravated if players are exposed to extended match play (10). Finally, degradation of the glycogen stores can reduce the glycolytic flux and increase the reliance on fat oxidation as well as exacerbate metabolic disruptions of the muscle and/or systemic homeostasis (10,11,12), which may provoke fatigue via peripheral or central mechanisms. Extended match play in soccer implies that players will be exposed to a total of two hours intense intermittent exercise, which we hypothesized would challenge depletion of single fibers and mixed muscle glycogen stores. Specifically, that

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in comparison to an ordinary 90-min soccer game, more players would reach glycogen levels below 250 mmol kg dw that has been proposed as critical for maintained muscle function and performance (10). For the first time with parallel measures of performance and physiological measures of muscle metabolism, we wanted to explore how extended match time affected fatigue with special focus on its proposed association with glycogen depletion.

A few studies have examined the physiological effects of extra time in soccer. For example, Field et al. (13) reported decreased mechanical efficiency and increased fat oxidation during extra time using a simulated soccer protocol. Others, using simulated models and both peripheral nerve and transcranial magnetic stimulation, showed an exacerbated neuromuscular fatigue response to extra "playing" time (14). Moreover, a systematic review based on 11 studies reported decreased technical and physical performance after a simulated 120-min soccer games (15), though the authors also concluded that there is currently limited knowledge on the impact of extra time on muscle metabolism and physiological responses. In fact, to date there are no studies of extra time using real soccer matches. Therefore, the limited information that exist so far regarding the physiology and metabolism of overtime comes from simulated soccer conditions and not actual soccer match play.

Along these lines, the purpose of this study was to investigate the physiological, biochemical and performance responses during a 120-min soccer game in well-trained competitive male players with the main emphasis on the impact of extra time in relation to muscle glycogen depletion. We hypothesized that the additional 30 min of playing time would markedly impair game performance, fatigue development and muscle physiological responses, in

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part due to exacerbated muscle glycogen depletion below the proposed 250 mmol·kg^{-1} d.w. and in both main muscle fiber types after the extra time period.

METHODS

Experimental design

This study is part of the 120-min project funded by the Union of European Football Associations (UEFA) to investigate the physiological, performance, recovery and psychological effects of extra time in competitive soccer using a randomized, repeated measures, crossover, double-blind design. The research design of the 120-min project is shown in Figure 1. In this study, we report the physiological, metabolic and performance effects of a 30-min extra-time period based on analysis of the pooled data obtained from the first two games of each trial [n=40 observations, i.e. two trials with 20 observations (players) per trial], because the diet intervention was not initiated at this point (supplementation took place after game 1), thereby allowing us to study only the net effect of extra time. Therefore, the first two games of the placebo and carbohydrate tria were performed under identical conditions. This project was pre-registered at ClinicalTrials.gov (ID: NCT04159194). Participants (n=20) underwent a thorough baseline screening including assessment of body composition, technical skills and physical performance, resting metabolic rate, daily dietary intake and physical activity level, followed by a familiarization period during which they were accustomed to the experimental procedures and participated in light training to develop team cohesion over a 7-day period. Subsequently, they completed two trials (placebo vs increased carbohydrate intake during the post-game recovery period) separated by a 10-day wash-out period. Each trial included two 120-min official games (two teams played against each other) performed 3 days apart according to UEFA official

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schedules for national team tournaments. Field activity (using GPS instrumentation), heart rate, technical performance and water consumption were monitored continuously during matches, as previously done in 90 min games (7). Body mass measurements (to estimate sweat loss), performance testing and blood sampling were performed before each game, at baseline, and after 90 and 120 min of play according to procedures used by multiple times by our research group (4-9). Muscle biopsies were collected before game 1 and 2 of each trial and at 90 and 120 min of game 1 of each trial. There were multiple biopsy stations performance testing stations to be able to perform the biopsies and testing as fast as possible. Players were taken out of the game in groups of five in order to approximately comply with the 5-min interval time allowed in official games following the game and before the initiation of extra time. Players were taken out of the game in the interval 87-94 and 117-124 min after the 90 and 120-min, respectively. During the testing they were replaced by substitute players and then entered the game immediately after testing. The rationale for applying broad-spectrum testing enable us to link game performance, fatigue development and both physiological alterations at both systemic and muscular level. Participants were asked to follow a standardized meal plan during the entire experimental period.

Participants

A total of n=27 competitive male soccer players (representing all outfield positions) were initially approached and underwent medical screening. The participants provided written consent to participate after they were fully informed about the experimental procedures and potential risks/discomforts associated with the study. The experimental procedures were in accordance with the World Medical Association Code of Ethics (Helsinki Declaration, as revised in 2013), and approval was obtained from the Institutional Ethics Committee (Protocol Number: 1462).

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Participation in the study was secured if participants: (i) had played at a competitive level (top three divisions) for ≥4 years; (ii) had practiced ≥5 times/week and participated in ≥1 game(s)/week in the last 5 seasons; (iii) were free of recent injuries and/or illnesses (≥8 months before the study); (iv) were not consuming ergogenic supplements/medications (≥8 months before the study); and (v) were not smokers. From the initial pool of 27 players, there were 25 outfield players that met the inclusion criteria whereas two players declined to participate in the study. Subsequently, 20 players were randomly selected and participated in the study (8 defenders, 6 midfielders and 6 attackers), while the remaining 5 players (not selected during randomization) were appointed as substitutes for the main participants (in the event of injury). The participants' physical characteristics are shown in Table 1. Thirty observations were included in the analysis (only cases without any missing measurement time-points were selected from the 40 observations that were examined in total in this study). Figure 2 provides the CONSORT diagram of the study. According to a power analysis (effect size of 0.3, power of 0.85, α probability error of 0.05), a sample size of 11 participants is required for a within-between factors repeated-measures analyses of variance with two trials and 3 measurement time-points (per trial).

Baseline measurements

Body mass and height were measured on a beam balance with a stadiometer (Beam Balance Stadiometer, SECA, Vogel & Halke, Hamburg, Germany), as previously described, and body mass index (BMI) was calculated (16). Body composition was assessed using dual-emission X-ray absorptiometry (GE Healthcare, Lunar DPX-NT), as previously described (17). Open-circuit spirometry was utilized for VO_{2max} assessments using an automated online

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pulmonary gas exchange system (Vmax Encore 29, BEBJO296, CareFusion, USA) via breath-by-breath analysis during a graded exercise test on a treadmill (Stex 8025 T, Korea), as described elsewhere (7). Soccer-specific conditioning was tested using the Yo-Yo intermittent endurance test level 2 (Yo-Yo IE2) and the Yo-Yo intermittent recovery test level 2 (Yo-Yo IR2) following standard procedures previously described (18). The Yo-Yo and VO_{2max} tests were performed on separate days. The participants' level of technical skill performance was evaluated using the creative speed and short dribbling tests, as described before (18).

Performance measurements

Countermovement jump (CMJ) height was measured on an Ergojump contact platform (Newtest Oy, Oulu, Finland) (16). The average of the three trials was calculated and used as the test result. To measure repeated sprint ability (RSA), the participants performed five consecutive 30-m sprints separated by 25-s of active recovery during which they jogged back to the starting line (5). Sprint times were measured with infrared photocells with a precision of 0.01 s (Newtest Oy, Oulu, Finland). Peak isometric force was determined at baseline and after 120 min in knee extensors and flexors in the dominant and non-dominant leg, as previously described (6). The CMJ and RSA testswere performed following biopsy sampling. The CMJ was always performed prior the RSA test.

Measurement of field activity pattern and hydration status

Field locomotor activity during match-play was monitored using a high-resolution global positioning system (GPS) (10 Hz, Viper Units; STATSports, Newry, Ireland) (19). Field activity was classified as total distance covered, mean speed, high-speed running (HSR, distance covered

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at speeds >21 km/h), sprinting (distance covered at speeds >24 km/h), number of high accelerations (>2 m/s) and number of high decelerations (>-2 m/s) (7). Intensity during match play was monitored using continuous heart-rate measurements (Team Polar Pro system, Polar, Kempele, Finland). Individual skill-related performance variables (rates of successful passes and duels won) were examined by experienced soccer analysts. To determine sweat loss during the 120-min match, players were weighed wearing dry shorts before each match and at 90 min and 120 min using a beam balance with a stadiometer as previously described (16). The water intake of each participant was recorded throughout the match.

Blood sampling and assays

Fasting blood samples were collected by venipuncture using a disposable needle (20-gauge) and a Vacutainer tube holder from an antecubital arm vein with the participants sitting. Plasma was prepared by centrifugation (1370 g, 4°C, 10 min) from blood samples collected into tubes containing ethylenediaminetetraacetic acid (EDTA) to measure glucose, glycerol and ammonia. All samples were stored at -80°C in multiple aliquots until assayed. A small portion (2 mL) of whole blood was collected into tubes containing EDTA to assess white blood cells (WBC), red blood cells (RBC), lymphocytes, monocytes, granulocytes, hemoglobin, and hematocrit using an automated hematology analyzer (Mythic 18, Orphee SA, Geneva, Switzerland). All samples were thawed only once before being analyzed and were protected from light and auto-oxidation. All assays were performed in duplicate, and samples collected after a match were corrected for plasma volume changes as described (20). Blood lactate concentration was measured using a hand-held portable analyzer (Lactate Plus; Nova Biomedical, Waltham, MA) as described (21). Plasma glucose concentration was determined using the Clinical

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Chemistry Analyzer Z1145 (Zafiropoulos Diagnostica S.A., Koropi, Greece) and a commercially available kit (Zafiropoulos Diagnostica S.A., Koropi, Greece). Glycerol concentration was determined by a commercially available kit (Glycerol Assay Kit, MAK117, Sigma-Aldrich®, St. Louis, IL, US) using a coupled enzyme assay involving glycerol kinase and glycerol phosphate oxidase, resulting in a colorimetric/fluorometric product proportional to the glycerol present. Plasma ammonia (NH_3) concentration was quantitatively determined on a Cobas Integra 400 plus (Roche Diagnostics, Rotkreuz, Switzerland) analyzer using a commercially available kit (NH_3L, 0-168, Cobas® substrates) in accordance with the manufacturer's instructions. Inter- and intra-assay coefficients of variation for the aforementioned variables ranged from 2.2 to 7.1% and from 3.2 to 7.8%, respectively.

Muscle biopsy sampling and assays

Muscle biopsies (~70-100 mg w.w.) were obtained from m. vastus lateralis in the dominant leg using the needle biopsy technique with suction, with the participants supine position. Prior to the game two incisions were made in the medial part of the muscle, as previously described (5,9), to be able to take the 90 and 120-min biopsies fast (within 30-40 s). The baseline biopsy was obtained from the more distal incision and the 90 and 120-min biopsies were collected from the more proximal incision. The muscle tissue was immediately frozen in liquid N_2 and stored at −80°C. The frozen sample was weighed before and after freeze-drying to determine water content. After freeze-drying, the muscle samples were dissected free of blood, fat and connective tissue. After extraction with $HClO_4$, neutralized extracts were analyzed for lactate, as previously described (22). Muscle glycogen content was determined spectrophotometrically (Beckman DU 650), as previously described (23). Freeze-dried muscle

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tissue (1.5 mg) was boiled in HCL (1 M, 0.5 mL) for 150 min before it was quickly cooled, whirl-mixed and centrifuged (3500 g, 10 min, 4°C). Boiled muscle sample (40 μL) and reagent solution (1 mL) containing Tris buffer (1 M), distilled water, ATP (100 mM), $MgCl_2$ (1 M), NADP+ (100 mM) and G-6-PDH were mixed before the process was initiated by adding diluted hexokinase (10 μL). Absorbance was recorded for 60 min before the glycogen content was calculated. Muscle glycogen was expressed as mmol/kg of dry weight. Analysis of glycogen content was performed on cross-sections (8 μm) cut from tissue-tec embedded muscle samples. A section was stained with Periodic acid–Schiff (*PAS*) stain and MHC Fast for visualization of glycogen content and FT myofibres. Individual myofibre relative glycogen content was based on the average glycogen density of the 25 most full (100%) and empty (0%) ST and FT myofibres, from which the fibre-type-specific relative glycogen content of individual myofibres was calculated (9).

Statistical analysis

Data normality was verified using the Shapiro-Wilk test. Because the vast majority of our data sets were normally distributed, we applied parametric statistics. A repeated-measures ANOVA was utilized to detect different time-point changes, accompanied by a Bonferonni *post-hoc* analysis test when a significant main effect for time was observed. For statistically meaningful differences, effect sizes (ES) and confidence intervals (CI) were calculated according to the corrected-for-bias Hedge's g method. ES was considered none, small, medium-sized and large for values 0.00-0.19, 0.20-0.49, 0.50-0.79 and ≥0.8, respectively. The between-day (baseline data of the first games 1 in trials 1 and 2) intra-individual coefficients of variation for the main outcome variables (muscle glycogen, muscle lactate, blood lactate, plasma ammonia,

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plasma glycerol, blood glucose, RSA fatigue index and CMJ performance) have been calculated [using the formula (standard deviation / mean) x 100] and incorporated into the results section. The level of statistical significance was set at $p<0.05$. Data are presented as means±SD. Analysis was performed using the SPSS 20.0 software (IBM SPSS Statistics, Armonk, NY, US).

RESULTS

Match locomotor activity, heart rate, sweat loss

During the 30-min extra-time period, total distance covered, sprinting and high-intensity running expressed as distance per min were 9.7% (p<0.01; ES=1.15 [0.60;1.70]), 0.8% (p<0.01; ES=0.01 [-0.50;0.51]) and 11.6% (p<0.01; ES=0.19 [-0.31;0.70]), respectively, lower compared to the 90-min normal-time period (Table 2). Also, average locomotion speed and maximal running speed were 10.9% (p<0.01; ES=1.06 [0.52;1.60]) and 10.1% (p<0.01; ES=1.51 [0.94;2.09]), respectively, lower in extra time compared to normal time (Table 2). Intense accelerations and decelerations per min were 18.3% (p<0.01; ES=0.69 [0.17;1.21]) and 16.8% (p<0.01; ES=0.70 [0.18;1.22]), respectively, lower in extra time than in normal time (Table 2). In order to compare the last part of normal time with extra time, the last 15-min period of normal time (75-90 min) was tested against the two 15-min periods of extra time (90-105 and 105-120 min). Total distance covered, high-intensity running and sprinting distance were 5-12, 19-36 and 27-47% lower (all p<0.05) in the two 15-min periods of extra time compared to the last 15-min period of normal time. Moreover, peak sprinting speed declined (p<0.05) from 31.1±1.6 $km \cdot h^{-1}$ in the 75-90-min period to 30.1±2.0 and 28.1±2.1 $km \cdot h^{-1}$ in the two 15-min periods of extra time. Finally, the frequency of intense accelerations and decelerations was lower (p<0.05) in extra time than at the end of normal time (data not shown). Successful passes and duels won also

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declined by 5.3% ($p<0.01$; ES=0.70 [0.18;1.22]) and 10.7% ($p<0.01$; ES=2.07 [1.44;2.69]), respectively, during extra time (Table 2). Average heart rate was 169±9 beats/min during the first 90 min, but declined to 163±8 beats/min during extra time ($p<0.01$; ES=0.74 [0.21;1.26]) (Table 2). The total sweat loss rate rose by 67.8% ($p<0.01$; ES=2.24 [-2.89; -1.59]) during extra time compared to the previous 90 min, resulting in a 25.9% higher ($p<0.01$; ES=0.99 [-1.53; -0.45]) sweat rate in extra time compared to normal time (Table 2).

Sprint ability, jump performance and muscle function

The between-day intra-coefficient of variation for RSA fatigue index and CMJ was 0.4% and 1.4%, respectively. The RSA fatigue index was increased by 1.7% at the end of the 90-min game compared to baseline ($p<0.01$; ES=16.8 [-19.8; -13.7]) and had risen a further 6.6% by the end of extra time ($p<0.01$; ES=4.94 [-5.95; -3.92]) (Fig. 3A). Likewise, CMJ performance declined by 19% and 27% at 90 min ($p<0.01$; ES=1.15 [0.60; 1.69]) and 120 min ($p<0.01$; ES=1.49 [0.92; 2.07]), respectively, compared to baseline, with CMJ performance at 120 min 11% lower ($p<0.01$; ES=0.45 [-0.06; 0.96]) compared to 90 min (Fig. 3B). Peak isometric force, as a marker of muscle function, declined ($p<0.05$) from baseline to 120 min. Knee extensor strength decreased from 3.9±0.6 and 3.9±0.6 to 3.6±0.6 and 3.6±0.6 Nm/kg in the dominant and non-dominate leg, respectively (Dominant: $p<0.05$; ES=0.55 [-0.18; 1.28], Non Dominant: $p<0.05$; ES=0.54 [-0.19; 1.26]) (Fig. 3C-D). Knee flexor strength declined from 2.6±0.3 and 2.6±0.2 to 2.3±0.3 and 2.3±0.2 Nm/kg in the dominant and non-dominate leg, respectively (Dominant: $p<0.05$; ES=1.65 [0.82; 2.48], Non Dominant: $p<0.05$; ES=1.36 [0.57; 2.16]) (Fig. 3E-F).

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Blood metabolite response

The between-day intra-coefficient of variation for blood glucose, plasma glycerol, ammonia, and blood lactate was, 3.6%, 8.2%, 7.3% and 14.1%, respectively. Blood glucose was unchanged at 90 min compared to baseline, but after the 30-min extra-time period there was a 13% reduction compared to baseline ($p<0.01$; ES=1.41 [0.85;1.98]) and 90 min ($p<0.01$; ES=1.40 [0.84;1.96]) (Fig. 4A). Plasma glycerol doubled from baseline to 90 min ($p<0.01$; ES=3.20 [-3.96; -2.43]), and at 120 min had further increased by 137% ($p< 0.01$; ES=5.26 [-=6.33; -4.19]) and 22% ($p<0.01$; ES=1.18 [-1.73; -0.63]) compared to baseline and 90 min, respectively (Fig. 4B). Likewise, plasma ammonia was increased both at 90 min (+53%, $p<0.01$; ES=3.36 [-4.14; -2.57]) and 120 min (+191%, $p<0.01$; ES=3.23 [-3.99; -2.46]), with a 90% higher concentration at 120 min compared to 90 min ($p<0.01$; ES=2.34 [-2.99; -1.68]) (Fig. 4C). Blood lactate had increased ~5-fold at 90 min ($p<0.01$; ES=6.41 [-7.67; -5.16]) and ~6-fold at 120 min ($p<0.01$; ES=8.61 [-10.24; -6.99]), with a significant effect of extra time compared to normal time (90 min: 4.4±0.7 vs 120 min: 5.6±0.7 mmol/l; $p<0.01$; ES=1.69 [-2.28; -1.10]) (Fig. 4D).

Muscle metabolic responses

The between-day intra-coefficient of variation for muscle lactate and glycogen was 19.5% and 11.1%, respectively. Muscle lactate increased by almost 2-fold and 4-fold at 90 min ($p<0.01$; ES=1.78 [-2.44; -1.13]) and 120 min ($p<0.01$; ES=1.66 [-2.30; -1.01]), respectively, compared to baseline, with extra time inducing a further rise of 103% compared to normal time ($p<0.01$; ES=1.13 [-1.73;-0.53]) (Fig. 4E). Muscle glycogen declined by 29% at 90 min ($p<0.01$; ES=1.70 [0.96;2.44]) and by 50% at 120 min ($p<0.01$; ES=3.20 [2.24;4.15]) compared to

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baseline (Fig. 5A). Extra time induced a further decline of 30% compared to 90 min (90 min: 266.1±64.0 mmol·kg^{-1} d.w. vs 120 min: 185.6±55.7 mmol·kg^{-1} d.w.; $p<0.01$; ES=1.31 [0.61;2.01]), with several individual values falling below the proposed critical level of 250 mmol·kg^{-1} d.w. (Fig. 5A). The muscle glycogen concentrations at 120 min correlated inversely (r=-0.56) to the fatigue index during the RSA test after the extra-time interval. The glycogen degradation rate almost doubled during extra time compared to the first 90 min ($p<0.05$; ES=0.83 [-1.49-0.17]) of the game (Fig. 5B). The analyses of the fiber-type-specific glycogen depletion pattern revealed that 69% and 66% of ST and FT fibres, respectively, were completely or party full of glycogen at baseline (Fig. 5C). At 90 min, this had dropped to 29% for ST ($p<0.01$; ES=2.10 [1.06;3.14]) and 37% for FT fibres ($p=0.05$; ES=1.29 [0.37;2.21]), and at 120 min it had further declined to 18% ($p<0.01$; ES=2.82 [1.64;4.01]) and 22% ($p<0.05$; ES=1.89 [0.88;2.89]) for ST and FT, respectively (Figure 5C), with a significant effect of extra time compared to normal time (ST: $p<0.05$; ES=0.96 [0.08;1.84]; FT: $p<0.05$; ES=0.75 [-0.11;1.62]). Accordingly, the percentage of ST and FT fibres that were completely or almost empty increased progressively after 90 min (ST: $p<0.01$; ES=2.10 [-3.14;-1.06] vs FT: $p=0.05$; ES=1.29 [-2.21;-0.37]) and 120 min (ST: $p<0.01$; ES=2.82 [-4.01;-1.64] vs. FT: $p<0.05$; ES=1.89 [-2.90;-0.89]) compared to baseline values, with a significant effect of extra time compared to normal time (ST: $p<0.05$; ES=0.96 [-1.84;-0.08]; FT: $p<0.05$; ES=0.76 [-1.62;0.11]). Specifically, 72% of ST and 64% of FT fibres after 90 min and 82% of ST and 78% of FT fibres after 120 min were completely or almost empty of glycogen compared to baseline (31% for ST and 35% for FT) (Fig. 5C).

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Hematological responses

Hematocrit had increased from baseline both at 90 min ($p<0.01$; ES=0.65 [-1.17; -0.13]) and 120 min ($p<0.01$; ES=0.71 [-1.23; -0.19]), without any effect of extra time, whereas hemoglobin concentration remained unaltered throughout the 120-min game (Table 3). Red blood cell count had increased at 90 min ($p<0.01$; ES=0.99 [-1.52; -0.45]) and subsequently declined ($p<0.01$; ES=0.49 [-0.02;1.01]) (Table 3). WBC, monocyte and granulocyte counts had increased at 90 min ($p<0.01$; WBC: ES=2.81 [-3.53; -2.10]; MON: ES=-2.36 [-3.01; -1.70]; GRAN: ES=2.66 [-3.36; -1.97]) and 120 min ($p<0.01$; WBC: ES=3.06 [-3.80; -2.31]; MON: ES=2.32 [-2.97; -1.66]; GRAN: ES=3.08 [-3.82; -2.33]) compared to baseline, without any effect of extra time (Table 3). No changes were noted for lymphocytes (Table 3).

DISCUSSION

The study is the first to investigate how extended match time affects muscle metabolic and physiological responses of importance for performance and fatigue development in well-trained competitive male soccer players. Our results demonstrate that the extended game time exacerbated glycogen depletion of single fibres, elevated fat breakdown, lowered blood glucose and induced hyperammonemia. These metabolic disturbances were associated with considerable deterioration of performance assessed as game distances covered, frequency of intense actions and impaired post-match muscle function, sprint ability and jump performance.

We found that physical game performance was markedly lower during the extra 30 min compared to the first 90 min. Indeed, both total ground covered and high-intensity running expressed as distance per min were 10-12% lower in the extra-time period. Also, sprint distance

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and maximal running speed, as well as acceleration and deceleration counts, declined. Finally, all of these physical performance markers were lower in extra time compared to the end of normal time. Thus, tracking parameters representing endurance, high-intensity performance and ballistic/explosive movements in competitive soccer (7, 12, 24) appeared to be impaired during extra time compared to normal time. This is supported by lower average locomotion speed and heart rates, also indicating that the general game intensity was compromised either by fatigue and/or a changed pacing strategy. Decrement in performance towards the end of a normal soccer game is a consistent finding in tracking research (25) and has been suggested to be caused by physiologically mediated fatigue (26). A recent systematic review confirmed our findings of decreased work rate in the extra-time period versus normal time (15) and reported a decrement in the same magnitude range (5-12%) as our study.

In support of our tracking data, a greater fatigue response was also observed after extra time compared to normal time in both RSA and CMJ performance (Fig. 3). Several studies have demonstrated that RSA is impaired after a 90-min game (4, 5, 7, 9, 27), while some (28-30), but not all (31, 32), report an effect on countermovement jump performance. Moreover, it was observed that peak isometric force had also deteriorated by approximately 10% both in knee extensors and flexors after extra time, which previously has been shown after normal games (6). Our findings above are supported by others reporting an exaggerated neuromuscular fatigue (14) response after extra time compared to normal time. Thus, the accumulated fatigue response observed at the end of a 90-min soccer game seems to progress substantially during an extra-time period in trained male soccer athletes.

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After the initial 90-min game duration, skeletal muscle glycogen stores were reduced to 266±64 mmol·kg^{-1} d.w., which is in line with other previous studies in soccer (4, 5, 9, 11, 33). More so, after the extra-time period a further drop to 186±56 mmol·kg^{-1} d.w. was evident corresponding to an approximate 50% reduction compared to baseline values. There appears to be a critical glycogen threshold around 250 mmol·kg^{-1} d.w. where high-intensity exercise performance is impaired by reductions in glycogen content (10, 34). Accordingly, the muscle glycogen concentration was just above this potential threshold after normal time, but declined markedly below this point during extra time which may explain the exacerbated fatigue response observed. Surprisingly, the muscle glycogen degradation rate was considerably faster during extra time than normal time, which is also indicative of high glycolytic activity during the final stages. This was unexpected since the amount of high-intensity activity was lower in extra time and a down-regulation of glycolysis is to be expected at this point due to the degraded glycogen levels with a concomitant up-regulated beta oxidation (9). However, importantly this may be biased by our experimental design with the repeated sprint test performed prior to the extra time period which have contributed to the reductions in muscle glycogen during the extra time period (35). Despite this, the glycogen utilization during the repeated sprint test is expected to be minor compared to the total load on the players during the complete 120-min game with several sprinting and high-intensity actions performed meaning that the end-exercise glycogen concentrations are likely only slightly underestimated by the addition of the repeated sprints. Collectively, very low muscle glycogen concentrations are reached after the extra-time period in soccer, which may evoke the greater fatigue response.

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The mediocre muscle glycogen stores measured in the muscle homogenate were further supported by the fibre-type-specific analysis demonstrating that 76% of all muscle fibres were low on glycogen after 120 min, a pattern evident in both ST and FT fibres. At the single-fibre level, a greater glycogen deficiency was observed after extra time versus normal time, which backs up the findings at whole-muscle level. The discovery of a large number of individual fibres being nominal on glycogen after prolonged high-intensity intermittent exercise is in line with previous data after a 90-min soccer game (5, 9), simulated soccer (33) and elite ice-hockey match play (34). Accordingly, despite that the glycogen stores at whole muscle level (measured in a muscle homogenate sample) are non-depleted, several individual fibers reach levels that may be critically low to sustain normal function. Depleted glycogen in a large fraction of myocytes or subcellular compartments has not yet been solidly coupled to a single fatiguing mechanism, but has been suggested to disturb several acts in the excitation-contraction (E-C) coupling, such as the Na^{+}-K^{+} ATPases, Ca^{2+} ATPases and myosin ATPases (10). The potentiated fatigue response during extra time in a soccer game may therefore be associated with depleted muscle glycogen, which in an integrated manner may compromise key steps in the E-C coupling. Previous studies have shown statistical associations between degraded muscle glycogen and the decline in RSA after a team-sport game (5, 9, 34), which we failed to shown in the resent study, which may relate to the heterogensous distribution of glycogen in the myocytes.

A common finding during prolonged exercise is an accelerated reliance on fat to fuel ATP resynthesis (36), and in parallel up-regulation of beta oxidation has been shown when muscle glycogen becomes low (37). In our study, plasma glycerol rose by nearly 20% during the final 30 min, supporting a greater reliance on fat in the energy yield. During prolonged intense exercise,

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such as a soccer game, plasma insulin declines and catecholamines are being released into the bloodstream (10), as has also been shown during the later stages of a soccer game (9, 37), which stimulates glucagon production and the release of glucose from the hepatic glycogen stores (38). Since the exercise intensity is lowered in the extra-time period (Table 2), the frequency of rest and low-intensity exercise periods is increased, which may elevate blood flow in the adipose tissue and therefore also the release of free fatty acids. A consistent finding in previous studies is that hypoglycaemia does not occur during a 90-min soccer game (4, 5, 9, 33), which again was confirmed by the 90-min measurement in this study. However, after 120 min a ~15% decrease was observed compared to the 90-min time-point, suggesting moderately lowered blood glycose levels, which may activate fatiguing mechanisms of central origin (39). A method to evaluate coordination skill performance during a game is to analyze technical performance or event data (40). In this study, the fraction of successful passes declined by >5% during extra time compared to normal time. This is supported by observations in a systematic review by Field et al (15) demonstrating a reduction in technical performance events such as shot speed, number of passes and number of dribbles during extra time. The same review provided evidence (15) that carbohydrate supplementation during extra time may partly counteract these effects, supporting the proposition that hypoglycaemia, low glycogen or muscle fatigue may also be affecting these variables in the extra time period in soccer.

Interestingly, plasma ammonia increased by ~60% at 90 min, which confirms what others have observed in a male soccer game (9), but was augmented even further (2-fold) at 120 min compared to baseline, which represented an additional 90% increase as an effect of extra time alone. Ammonia is produced in the exercising muscles during energetically demanding work as

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result of purine nucleotide degradation, potentially in combination with potential amino acid catabolism (41). Ammonia has been substantiated to rise markedly during soccer-specific high-intensity intermittent exercise (9), and has been observed to cross the blood-brain barrier and accumulate in the cerebrospinal fluid during prolonged exercise (39). Since the brain has no effective uric cycle, ammonia may therefore disturb the neurotransmitter homeostasis in the brain and cause cerebral functions to deteriorate (42), which also may affect technical and physical performance. Thus, the extensive muscle and perhaps also liver glycogen depletion and blood glucose usage may provoke both mild-to-moderate hypoglycaemia and hyperammonemia, which can be speculated to cause fatigue mediated centrally.

The total sweat rate was 2.04±0.57 L during normal time, which is within the range usually observed in other studies at moderately high temperature (30), but rose to nearly 3.5 L during extra time, resulting in a 26% higher sweat rate in extra time compared to normal time. Despite the fact that no difference was observed in hematocrit levels between the 90- and 120-min time-points, dehydration may become a component contributing to deteriorated performance in the extra-time period, especially since a high number of elite soccer players are hyperhydrated in normal conditions (43).

A limitation in the study may be that the player knew beforehand that the game was going into extra-time, which may have caused them to pace accordingly. However, the tracking data demonstrate a work rate very similar to real competitive games (37). Moreover, the degree of fatigue and the physiological response at 90-min is nearly identical to the literature (3-8). Thus,

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based on these observations, the 90-min game load is likely to be comparable to a real competitive game.

CONCLUSIONS

In conclusion, our study demonstrates that the degree of fatigue after an extra-time period in competitive male soccer is exaggerated compared to a normal game, and this may be induced by low glycogen concentrations in a large portion of muscle fibres. Other factors facilitated by lowered blood glucose, hyperammonemia and/or dehydration may contribute to the deterioration in performance, which points to an integrated and multifaceted fatigue scenario.

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The results of this study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine. The authors report no conflicts of interest.

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FIGURE LEGENDS

Figure 1. Experimental design of the study.

Figure 2. CONSORT flow diagram of the study.

Figure 3. Repeated sprint ability expressed as a fatigue index (A) and countermovement jump performance (B) during the 120-min game protocol (n=30). * Indicates significant difference to baseline; # Indicates significant difference compared to 90 min.

Figure 4. Blood glucose (n=30) (A), plasma glycerol (n=30) (B), plasma ammonia (n=30) (C), blood lactate (n=30) (D) and muscle lactate (n=25) (E) concentrations at baseline, after normal time (90 min) and after extra time (120 min). * Indicates significant difference compared to baseline; # Indicates significant difference compared to 90 min.

Figure 5. Muscle glycogen concentration at whole-muscle level (n=19) (A), muscle glycogen degradation rate (n=19) (B) and fibre-type-specific glycogen depletion pattern (n=11) (C) during the 120-min game protocol. * Indicates significant difference compared to baseline; # Indicates significant difference compared to 90 min.

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Figure 1

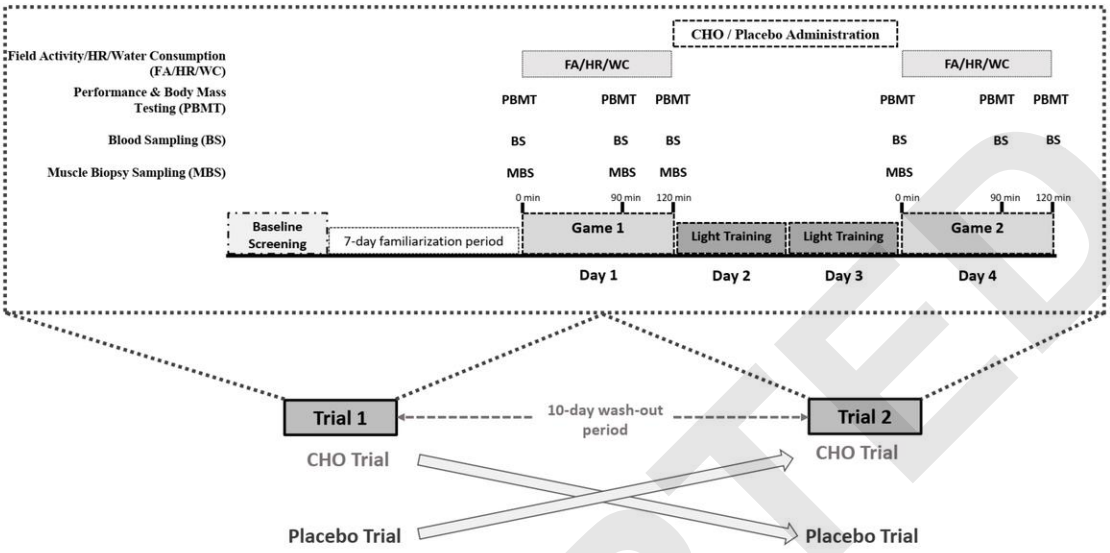

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Figure 2

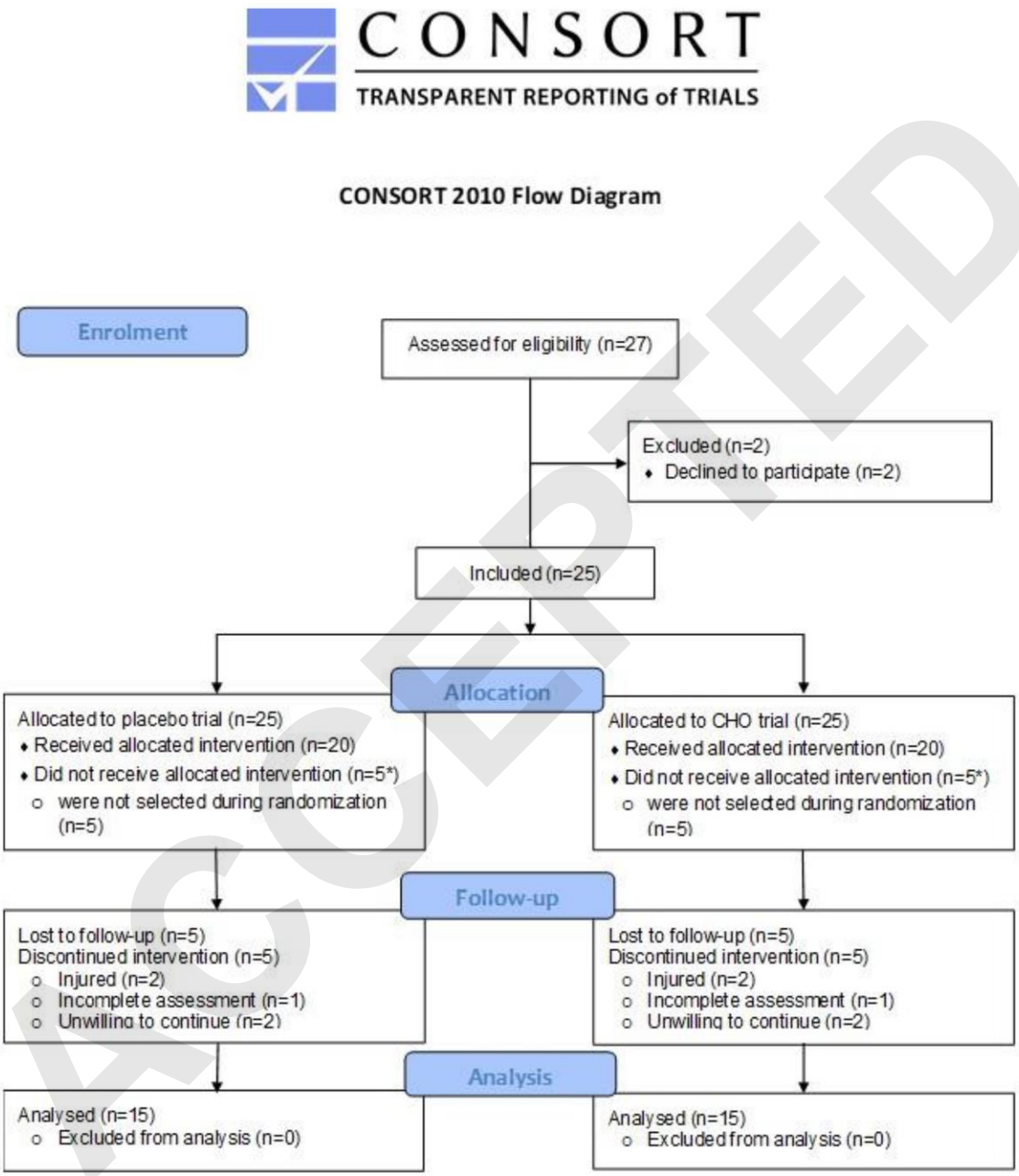

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Figure 3

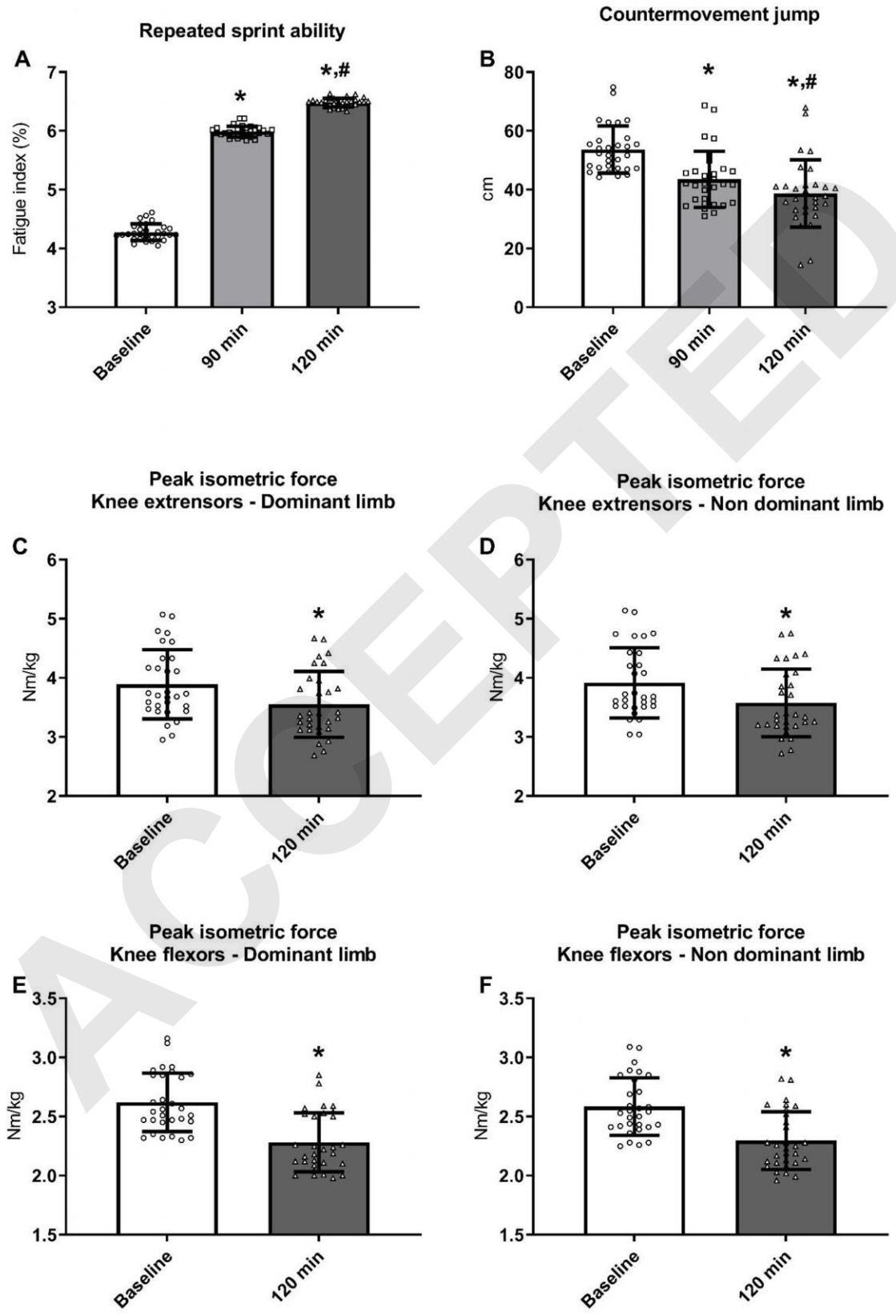

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Figure 4

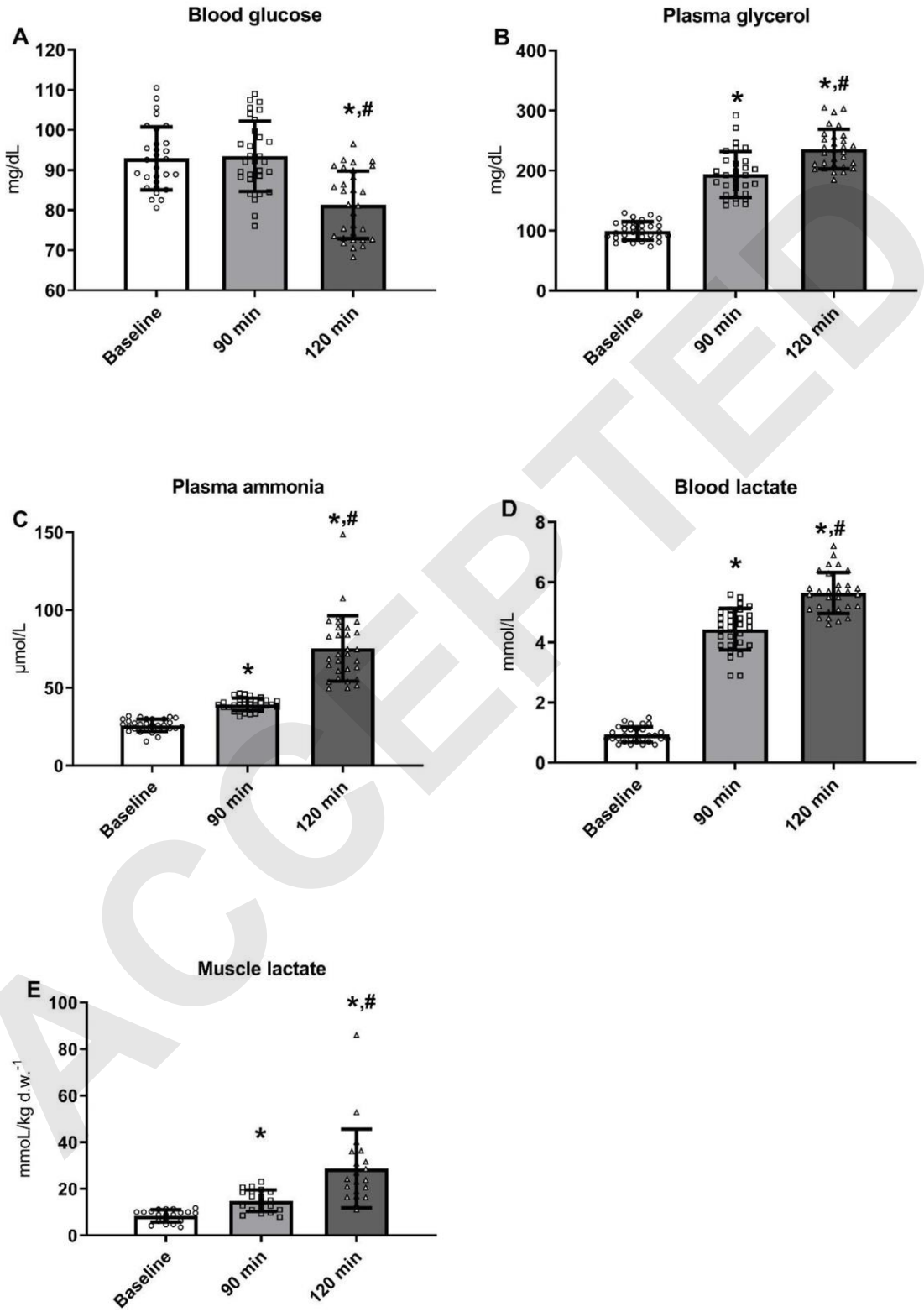

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Figure 5

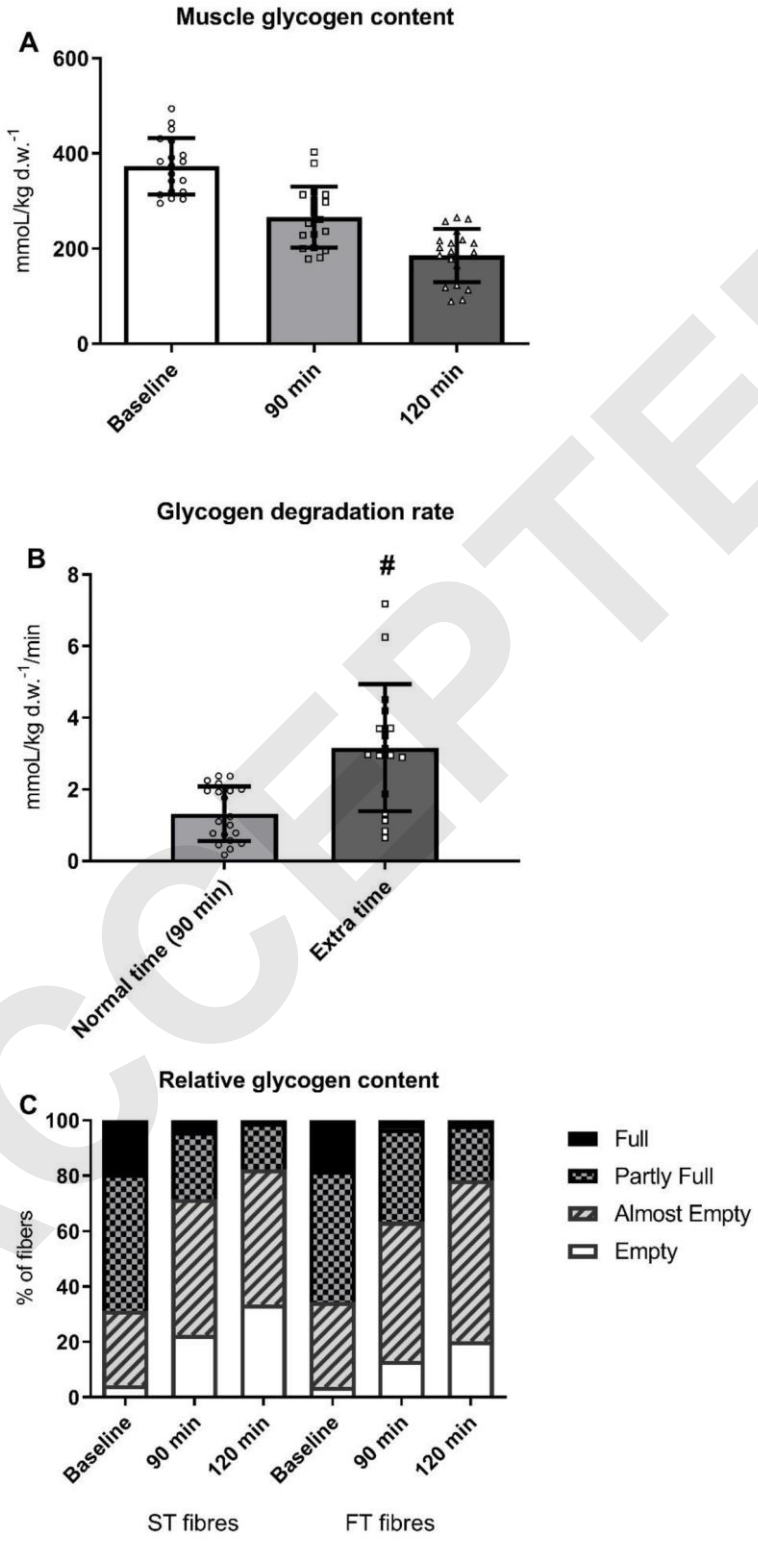

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Table 1. Participants' characteristics at baseline (n=20).

Variable	Mean ± SD
Age (yrs)	20.1 ± 1.4
Body height (m)	1.76 ± 0.06
Body mass (kg)	69.6 ± 4.7
BMI (kg/m^2)	22.4 ± 0.9
Fat (%)	14.9 ± 5.1
VO_2max (ml/kg/min)	59.4 ± 3.7
Maximum speed (km/h)	19.3 ± 1.3
HRmax (b/min)	200.1 ± 3.7
Resting HR (b/min)	61.4 ± 4.8
Lactate threshold (%)	79.5 ± 3.2
Yo-Yo IE2	
Speed level	18.7 ± 0.7
Distance (m)	3144 ± 350
HRmax (b/min)	196.0 ± 3.0
Yo-Yo IR2	
Speed level	23.2 ± 0.5
Distance (m)	1517 ± 168
HRmax (b/min)	196.6 ± 3.6
Skill level	
Creative speed test (s)	16.3 ± 0.7
Short dribbling test (s)	12.2 ± 0.6

BMI, body mass index; LBM, lean body mass; VO_2max, maximal oxygen uptake; HR: heart rate; Yo-Yo IE2, Yo-Yo intermittent endurance test level 2; Yo-Yo IR2, Yo-Yo intermittent recovery test level 2.

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Table 2. Match performance, HR and sweat loss during the 90-min match (0-90 min) and overtime (90-120 min).

	0-90 min	90-120 min	% change
Match Performance			
Total distance (m/min)	121.5 ± 9.4	109.7 ± 10.8#	-9.7
Sprinting (m/min)	1.29 ± 1.08	1.28 ± 1.19	-0.8
HIR (m/min)	2.32 ± 1.35	2.05 ± 1.4	-11.6
Average speed (km/h)	7.3 ± 0.6	6.6 ± 0.7#	-10.9
Maximal speed (km/h)	32.4 ± 2.3	29.1 ± 2.0#	-10.1
ACC/min	1.15 ± 0.3	0.94 ± 0.3#	-18.3
DEC/min	1.07 ± 0.2	0.89 ± 0.3#	-16.8
Successful passes (%)	71.4 ± 7.4	65.8 ± 8.4#	-5.3
Duels won (%)	45.4 ± 4.8	34.7 ± 5.4#	-10.7
Heart Rate			
Mean HR (beats/min)	168.8 ± 8.6	162.7 ± 7.7#	-3.6
Sweat Loss			
Sweat loss (mL)	2044.7 ± 574.3	3431.7 ± 645.6#	+67.8
Sweat rate (mL/min)	22.7 ± 6.4	28.6 ± 5.3#	+25.9

HR, heart rate; HIR, high intensity running; ACC, accelerations (> 2 m/s^2); DEC, decelerations (> 2 m/s^2); # Indicates significant difference with 0-90 min ($p < 0.05$).

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Table 3. WBC, RBC, hematocrit and haemoglobin responses at baseline, after the match (90 min) and after overtime (120 min).

	Baseline	90 min	120 min
WBC (10^3/uL)	6.0 ± 1.2	12.7 ± 3.1*	14.1 ± 3.5*
Lymphocytes (10^3/uL)	2.4 ± 0.4	2.6 ± 0.7	2.6 ± 0.6
Monocytes (10^3/uL)	0.26 ± 0.07	0.48 ± 0.11*	0.55 ± 0.16*
Granulocytes (10^3/uL)	3.4 ± 1.0	9.8 ± 3.2*	11.0 ± 3.3*
Hematocrit (%)	43.7 ± 1.6	44.9 ± 2.0*	45.0 ± 2.0*
Haemoglobin (g/dL)	15.1 ± 0.7	15.3 ± 0.9	15.2 ± 0.9
Red Blood Cells (10^6/uL)	4.9 ± 0.2	5.1 ± 0.2*	5.0 ± 0.2#

* Indicates significant difference with Baseline; # Indicates significant difference with 90min ($p < 0.05$).

ACCEPTED

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